Report on Practical 2

Fiber Optic Splicing

1. Aims

- Understand the steps and tools required for splicing Fiber optic cables.

2. Objectives

- Learn the preparation and alignment process for fiber optic cables.
- How to do fusion splicing.
- Practice using splicing machines for accurate splicing.
- Test the spliced fiber to ensure optimal performance.

3. Background

- Splice is when joining the ends of two optic fibers together
- It is a precise process critical for high-speed communication networks, requiring minimal signal loss and high efficiency.
- Two types of splicing are commonly used:

  -fusion splicing (melting two fibers together to create one single fiber) and

  -mechanical splicing (aligning and securing two fibers in a correct position)

  As we will be working with glass fiber it can crack and as the pieces are too small it could get on your skin and cause infection. Therefore, safety gloves and glasses are recommended

4. Procedure

1. Prepare Tools and Materials:

   Fusion splicer

   Cleaver

   Cutters and Strippers

Cleaning wipes

Heat-shrink protective sleeve

Fiber optic cables.

2. Add heat-shrink protective sleeve to the fiber first.
3. Strip the cable's outer coating using the stripper, exposing the fiber. And clean it thoroughly to remove dust and debris.
4. (important part) Use a cleaver to cut the fiber ends precisely for proper alignment. And refrain from touching afterwards as it may damage or dirty the clean cut which was just made.
5. Insert both the fibers into the fusion splicer machine (one from left and other right). Initiate the fusion splicer, which heats and joins the fibers seamlessly.
6. After splice is completed without any errors, gently slide a protective sleeve we put on at first over the splice to keep it secured.
7. Use an optical power meter or OTDR to verify the splicing quality.

5. Conclusion

Splicing is a cost effective, efficient way to join two fiber optic cables together. The process of fiber optic splicing involves careful stripping and cleaning the cable, using cleaver to cut the fiber ends, initiate the fusion splicer, and testing to ensure low-loss connections. By following the above procedures and using the appropriate tools, we achieve our output enabling efficient data transmission in fiber optic networks.

6. References

- YouTube video: Fiber Optic Splicing Guide & Demo https://www.youtube.com/watch?v=xba2MThR9Ls